

Omar Alkhatat

Software Engineer: Backend

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SUMMARY

Backend Software Engineer and Technical Team Lead with 3+ years of experience architecting scalable Python/Django systems while guiding development teams from concept to deployment. Proven track record in leading cross-functional projects—including banking, AI, and real-time tracking platforms—delivering robust APIs, clean architecture, and on-time results. Passionate about mentoring engineers, driving technical excellence, and building solutions that balance performance, maintainability, and business impact.

EDUCATION

- B.Sc. in Computer Science, Antioch Syrian University (2021-2026) — Graduated

LANGUAGES

- Arabic: native
- English: Professional Working Proficiency

COURSES

- Software Engineering — IBM
- Python for Data Science, AI & Development — IBM
- Linux Commands and Shell Scripting — IBM
- Git version control — IBM
- Django Application Development with SQL and Databases — IBM
- Developing AI Applications with Python and Flask — IBM
- Web Application Technologies and Django — Michigan University
- Computer skills & Communication skills — University

TECHNICAL SKILLS

- Programming Languages:** Python, C++, HTML, JavaScript
- Web Development:** Django, Django REST Framework, REST APIs
- AI & Machine Learning:** Neural Networks, Deep Reinforcement Learning (PPO, IPPO/MAPPO), RAG pipelines, LLM integration & multi-model routing, embeddings & vector retrieval, Genetic Algorithms, PyTorch, scikit-learn
- Database Management:** MySQL, SQL Server, SQLite, Elasticsearch
- Tools & Platforms:** Git, Linux, Windows, Streamlit
- Core Concepts:** Object-Oriented Programming (OOP), Data Structures, Requirements Analysis, Back-end Development

PROFESSIONAL SKILLS

- Project Management
- Leadership & Teamwork
- Creative Problem-Solving
- Strong Communication & Collaboration
- Organizational Skills

ACHIEVEMENTS

- Successfully managed and delivered multiple solo and team projects
- Recognized for creative thinking and innovative problem-solving in university projects

PROJECTS

Multi-Agent Reinforcement Learning for Adaptive Traffic Signal Control

Graduation project · Team of 2

Reinforcement-learning traffic controller coordinating nine signalised junctions in a SUMO simulation. Implemented IPPO and MAPPO from scratch with a continuous green-duration policy and a multi-term reward over queues, delay, throughput and spillback, then benchmarked both across 100 paired seeds and fine-tuned the winner onto two Damascus road networks.

Tools: Python, PyTorch, SUMO/TraCI, PettingZoo, Gymnasium, NumPy, pandas

MTNSYR — Telecom Data Warehouse & Analytics Platform

Team of 2

Analytics platform unifying MTN Syria and Syriatel subscriber data. Built the ETL that reconciles both schemas into a five-table star schema with incremental and full-reload modes, then a RandomForest customer-lifetime-value model and a spend-segment classifier, surfaced through a nine-page dashboard behind a single CLI.

Tools: Python, pandas, scikit-learn, SQLite, Streamlit, Plotly

Banking Management System

Team Lead

Engineered a comprehensive banking management system encompassing employee, customer, loan, and branch operations. Spearheaded development as Team Lead, taking full ownership of architecting, coding, and deploying the Branch and Employee Management modules from the ground up.

AI Chat Assistant

Architected a modular Python AI Chat Assistant featuring dual Rich terminal and Tkinter GUI interfaces, multi-LLM routing, persistent session storage, and intelligent rate-limiting with retry mechanisms; developed the core LLM abstraction layer and delivered a polished, production-ready frontend and backend integration.

Tools: Python, Rich, Tkinter, LLM APIs

Document Search Engine

Designed a scalable document search engine capable of automated content ingestion, efficient data indexing, and rapid information retrieval. Owned the complete backend implementation, engineering Django-Elasticsearch integration, developing robust document parsing pipelines, and optimizing index lifecycle management.

Tools: Python, Django, Elasticsearch

Real-Time Fleet Management Platform

Developed a real-time fleet management platform for continuous vehicle telemetry and geospatial monitoring. Solely engineered the entire backend infrastructure, implementing low-latency data processing, scalable API architecture, and secure tracking service endpoints.

Tools: Python, Django REST Framework, REST APIs

HIV Therapy Optimisation with Genetic Algorithms

Engineered an AI-driven clinical decision support system for optimizing HIV antiretroviral therapy protocols using Genetic Algorithms, dynamically balancing clinical efficacy, toxicity, cost, and patient adherence per WHO 2024 guidelines. Spearheaded end-to-end development, architecting a multi-objective fitness engine, integrating Stanford HIVDB resistance mapping, implementing patient-specific constraint validation, and deploying automated clinical reporting modules tailored for resource-limited healthcare environments.

Tools: Python, Genetic Algorithms, Stanford HIVDB

Global Weather Tracking Application

Built a global weather tracking application enabling users to curate personalized city lists and access real-time meteorological forecasts worldwide. Directed full-stack development, integrating external weather APIs, implementing intelligent caching strategies, and delivering a responsive, location-aware interface.

Tools: Python, Django, REST APIs